

Harrison Hoyos Plazas

AI ENGINEER | BACKEND ENGINEER | DATA SCIENTIST

Medellín, Colombia | +57 311-508-4628 | harrihoyos2680@gmail.com
linkedin.com/in/harrison194 | github.com/Wiz80 | platzi.com/p/harry-hoyos

PROFILE

AI Engineer with 5+ years of experience building production-grade LLM systems, RAG architectures, and Python backends. Deep expertise in LangChain & LangGraph, vector databases, semantic search pipelines, and structured prompt engineering. Proven track record delivering scalable AI services across financial, medical, and real-estate domains. Skilled in MLOps, Docker/Kubernetes, and cloud infrastructure (AWS, GCP, Azure). Fluent English speaker with a strong engineering discipline (clean code, testing, CI/CD).

PROFESSIONAL EXPERIENCE

AI Engineer
Oct 2025 – Present

StrategyCorps

- Architected AskIQ, a multi-tenant LangGraph analytics agent run as a single Databricks App for all institution clients, with per-request tenant routing via a Unity Catalog registry — replacing a prior one-endpoint-per-client deployment model.
- Built a dual-index retrieval layer (custom Vector Search MCP tools over dashboard-widget semantics + gold/silver/staging financial models, paired with Databricks SQL) powering natural-language Q&A over 300+ dashboard metrics.
- Implemented per-tenant conversational memory on Lakebase (Postgres) and a production feedback loop (thumbs up/down → MLflow assessments → nightly re-evaluation).
- Designed a 7-stage LLM evaluation system: deterministic SQL-grounded scoring, MLflow LLM-judge scoring, judge alignment, and automated benchmarking with per-model drift detection/alerting.
- Built a closed-loop self-improvement pipeline ("Agent Doctor") that clusters eval failures and gates prompt-edit proposals through shadow evaluation + CI/CD auto-merge with rollback.

AI Engineer
May 2024 – Sep 2025

Capital Certainty
Full-time: May 2024 – Jan 2025 | Part-time: Feb 2025 – Sep 2025

- Developed RAG pipelines and hybrid retrieval combining semantic search, Elasticsearch BM25, and cross-encoder reranking to retrieve relevant context.
- Built multilingual document processing with language detection and translation to support retrieval across languages.
- Fine-tuned a 32B DeepSeek model from medical data resources, optimized using QLoRA, and developed specialized agents that provide contextual information about blood, urine, and genetic test results.
- Deployed and served models on company-managed cloud servers with NVIDIA A100 GPUs, using vLLM and Ollama. Applied tensor parallelism to optimize model inference.

AI Software Engineer
Feb 2025 – Sep 2025

FinSync
Part-time

- Built an event-driven KYA/KYB/KYC verification service (FastAPI + Celery + RabbitMQ) integrating Alloy's API; handled webhook callbacks and posted status/alerts to a Slack channel.
- Generated AI-driven Business Plan & Business Profile documents via guided prompts to capture context and produce structured deliverables.
- Automated financial health reporting (Income Statement, Balance Sheet, Cash Flow); enriched with client business-plan context and external knowledge in Qdrant (via LangChain) to deliver goal-oriented recommendations.
- Implemented a product-insights pipeline: MCP server integrated with RabbitMQ & FullStory, plus LlamaIndex (RAG agents) to surface user bottlenecks and improvement opportunities.

Data Scientist & DevOps Engineer
Feb 2023 – Apr 2024

Instituto Internacional de Ciencia de Datos

- Developed a specialized chatbot using Llama 2 and LangChain connected through a RESTful API built in FastAPI, deployed on AWS EC2, to answer business inquiries based on client databases.
- Implemented an MLOps pipeline for predicting photovoltaic generation in a Spanish solar park, utilizing Azure services for the complete ML lifecycle.
- Designed and implemented a fraud detection system for banking transactions using Azure MLOps tools, significantly improving accuracy and efficiency.

PROFESSIONAL EXPERIENCE / CONTINUED

Senior Python Developer / Data Scientist
Jan 2022 – Jan 2023

VIP Alliance

- Spearheaded the development of an API (FastAPI) hosting financial recommendation models, integrating complex business logic to provide accurate portfolio movement recommendations.
- We used Django to create APIs that enable seamless communication with an Apache Superset dashboard, facilitating automatic monthly updates of client recommendations.
- Integrated Interactive Brokers to manage individual client portfolios, implementing algorithmic trading based on monthly recommendations, enhancing operational efficiency by 30%.

Mid Python Developer
Jul 2021 – Jan 2022

FincaRaíz

- Led the integration and migration of data and APIs from a Python/Django backend to a PHP/Laravel backend for 360 Latam, a conglomerate of major Latin American real estate listing companies.
- Ensured seamless functioning of diverse business logics, facilitating the complementary operation of services provided by FincaRaíz and Infocasas.

Semi-Senior Python Developer / DevOps Engineer
Oct 2020 – Jul 2021

Imagine Apps – FincaRaíz client

Python: Oct 2020 – Apr 2021 (7 months) | DevOps: May 2021 – Jul 2021 (3 months)

- Led the development of a branch-based real estate business model, enhancing productivity and information distribution for clients.
- Created an endpoint for data consumption to display statistics and market insights for real estate agents and builders.
- Orchestrated the deployment of backend and frontend applications following Azure's CI/CD process across development, QA, and production environments.

DevOps Engineer Intern
Apr 2020 – Sep 2020

IBM

- Developed an integration of AWS tools (S3, PostgreSQL, Amazon Athena) with Power BI to create an insights dashboard for IBM Sterling team's client deliveries.
- Participated in a DevOps project to define a CI/CD pipeline for a web application deployment on Kubernetes in OpenShift, using Jenkins for automation.

TECHNICAL & SOFT SKILLS

Generative AI & Language Systems LangChain / LlamaIndex / ChromaDB; OpenAI / Anthropic / Gemini / Hugging Face; RAG Systems / Vector Databases; Semantic Search / BM25 / Cross-encoders

Data Analysis Tools & Frameworks TensorFlow / Keras / Scikit-Learn; Azure Machine Learning / SageMaker; Databricks / Spark / Hadoop; Power BI / Tableau / Apache Superset

MLOps & Model Management MLflow / DVC / Weight & Biases; Model Serving / Model Monitoring; Feature Store / Experiment Tracking

Programming & Scripting Python (Data Science, Backend); JavaScript; SQL; Bash

Web Development & APIs Django / FastAPI / Flask; API Gateway / OpenAPI / GraphQL; Nginx / Apache / Load Balancing

Cloud & Infrastructure Azure (AKS, Functions, ML Services); AWS (ECS, Lambda, SageMaker); GCP (Vertex AI, Compute Engine, BigQuery); Terraform / CloudFormation / Ansible

Database & Storage PostgreSQL / MySQL / Azure SQL; MongoDB / DynamoDB / Cassandra; Data Lakes / ETL Pipelines

DevOps & CI/CD Docker / Kubernetes / Container Orchestration; GitHub Actions / GitLab CI / Azure DevOps; Infrastructure as Code / DevOps Practices

Monitoring & Observability Grafana / Prometheus / CloudWatch; ELK Stack / Datadog / APM; Metrics / Logging / Alerting

Leadership & Project Management Technical Leadership / Mentoring; Project Planning / Risk Management; Stakeholder Communication; Cross-functional Team Collaboration

EDUCATION & LANGUAGES

Universidad Industrial de Santander (UIS)
Jul 2017 – Jun 2023

Electrical Engineering | Bucaramanga, Colombia

School of Electrical, Electronic and Telecommunications Engineering (E3T)

English – B2+: Advanced Languages School UIS / Other courses (2018 – Present)
Français – B1: Intermediate Languages School UIS (2019 – 2021)